Cost, Carbon, and the Speed of Replenishment in Distribution Network Design

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Abstract

A firm that distributes through regional distribution centers decides how many to operate and how fast to replenish them. Each decision trades one source of emissions against another: slower line haul burns less fuel but requires more safety stock, and more distribution centers shorten the last mile but add buildings and give up the pooling that keeps inventory low. We model the two decisions jointly under a carbon price or cap and show that the design problem has a single best solution at every carbon price. Which way carbon pricing moves the network, faster or slower and denser or more consolidated, follows from a comparison of carbon intensity per dollar across holding, line-haul freight, last-mile delivery and facilities, weighted by the share of a distribution center's cost that is lost pooling. Calibrated to public US emission factors and logistics costs, the comparison turns on how speed is bought and how the last mile is driven. Speed bought with fuel, through faster vessels, truck instead of intermodal rail, or air, emits more per dollar than holding the stock of every product we examine, perishables included, so carbon pricing slows it; speed bought with labor, such as team drivers, adds little carbon and is favored. Heavy diesel delivery favors more distribution centers, and electric vans fewer.

Keywords: Logistics; Distribution network design; Carbon emissions; Safety stock; Replenishment speed

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1 Introduction

Distribution networks are being redesigned on two fronts. Firms are moving stock closer to demand: Amazon divided its US fulfillment network into eight regions and reported a 19% reduction in the miles traveled to deliver packages (Amazon.com, Inc. 2023), and Home Depot is building flatbed distribution centers in its forty largest markets (The Home Depot 2020). Firms are also slowing their line haul: AstraZeneca has moved most of its international freight from air to ocean and ships about 65% of its cargo by sea (Pharmaceutical Commerce 2023), and truckload shippers typically convert lanes to intermodal rail once the savings reach 10 to 15% (FreightWaves 2026). These moves are made mainly for cost and service, but carbon is now entering them directly. The European Union has priced the carbon of maritime shipping since 2024 (European Parliament and Council of the European Union 2023) and will price road fuel from 2028 (European Environment Agency 2026), California is phasing in the reporting of supply-chain emissions from 2027 (State of California 2023), and ISO 14083 standardizes how the emissions of transport and warehouses are counted (International Organization for Standardization 2023). A firm that prices or caps its distribution carbon must therefore ask which way carbon pushes two basic design decisions: how many distribution centers to operate, and how fast to replenish them.

A firm serves its markets from a network of distribution centers that it replenishes over time. Among the operational choices this involves, we study two: the *density* of the network — how many distribution centers the firm operates — and the *speed* with which those centers are replenished. Both shape the firm's cost, and both shape the carbon its distribution emits. With carbon priced and reported (McKinnon and Piecyk 2012), a firm sets both with carbon as well as cost in view, and we study how a carbon price shapes that choice.

Each decision carries a standard cost–carbon trade-off. Network density governs proximity to markets: operating more distribution centers shortens the last-mile distance and changes the warehouse energy the network consumes. Replenishment speed governs availability: replenishing faster shortens the lead time that inventory must cover and changes the emissions of the freight that serves the network. Distribution emits carbon both when goods move and when they are held, and each affects both the firm's cost and its carbon. Cost balances these effects one way; with a price on carbon, the carbon-minimizing balance can fall elsewhere, and how it does so is the central question this paper addresses.

Slowing replenishment and consolidating the network are the common transport-side responses: slowing moves freight by slower, lower-emission modes, as in slow steaming in shipping (Psaraftis and Kontovas 2013; Cariou 2011), and consolidating serves markets from fewer, larger distribution centers (McKinnon and Piecyk 2012). Each response also affects inventory. A unit held in a warehouse consumes energy — refrigeration, climate control, handling — for as long as it is held. Slowing replenishment lengthens the lead time that a stock buffer must cover, so the safety stock, and the carbon of holding it, rises with the square root of that lead time; consolidating pools demand across markets (Eppen 1979) and saves facility energy while lengthening the last mile. Each substitution thus lowers one source of carbon and raises another, and whether it lowers the

total depends on which source is larger.

The choice between the two sources is a choice between two operational substitutions. A firm provides *availability* either by holding safety stock or by replenishing faster — a shorter lead time leaves less for the buffer to cover, so speed and inventory are substitutes for the same service — and it provides market *proximity* either by operating more local facilities or by delivering farther from fewer, so facilities and last-mile travel are substitutes for the same access. The cost-minimizing configuration uses one mix of these activities; when carbon is priced, the optimal mix can differ. When safety stock is more carbon-intensive per dollar than the inbound transport that would replace it, the carbon-minimizing chain provides availability through *faster* replenishment. When last-mile travel is more carbon-intensive per dollar than the facilities that would shorten it, the carbon-minimizing chain provides proximity through facilities — a *more dispersed* network. The lowest-carbon distribution system thus depends on the product and, as we show, even more on how the firm buys speed and how it delivers the last mile.

We formalize this in a single-product location–inventory model with stochastic demand. The firm chooses network density and replenishment speed to minimize cost subject to a carbon cap and a service-level requirement. Cost and carbon each comprise the same four components — facilities, holding, inbound transport, and last-mile transport. The model follows the location–inventory tradition (Daskin et al. 2002; Shen et al. 2003) in a continuous-approximation form (Daganzo 2005), with a distinguishing addition: replenishment speed is a decision. Because speed sets the lead time that stochastic safety stock must cover, it directly determines inventory carbon and couples the four components together.

The paper makes three contributions. First, it characterizes the joint design. The priced problem has a single best design at every carbon price, although it is not jointly convex, and the carbon-minimizing direction of both decisions follows from four carbon-intensity-per-dollar ratios and one cost share that the cost-minimizing design reveals: the share of a distribution center's marginal cost that is lost pooling. Because faster replenishment recovers part of the pooling that dispersion gives up, the carbon intensity of line-haul freight enters the choice of how many distribution centers to operate, a link that the analysis of either decision alone does not reveal. Second, it measures the ratios. From public US emission factors and logistics costs, activities whose cost is mostly energy emit far more carbon per dollar than those whose cost is mostly labor or capital. The classification therefore turns on how speed is bought and how the last mile is driven: speed bought with fuel is slowed under carbon pricing for every product we examine, perishables included; speed bought with labor is favored; heavy diesel delivery calls for more distribution centers, and electric vans for fewer. Third, it shows how far the design moves. The response to a carbon price grows with the share of carbon in each activity's priced cost, which is small at prices in current use: in a calibrated national network, a price of \$100 per tonne changes network carbon by less than one percent for ambient goods, whereas a cap at the carbon minimum triples the number of distribution centers when stores are served by diesel trucks. A firm can compute the diagnostic from its own rate data, and the same ratios give the carbon price at which carbon

becomes a material share of each activity's cost.

The remainder of the paper proceeds as follows. Section 2 positions the paper relative to the location–inventory, carbon-network-design, and inventory–transport–emissions literatures. Section 3 develops the model, Section 4 establishes the results, Section 5 applies the results to calibrated US data, Section 6 draws out the managerial implications, and Section 7 concludes.

2 Related Literature

This study connects four streams: location–inventory design with risk pooling, carbon-aware network design, the joint control of inventory and transport mode under emissions regulation, and carbon as a priced operational cost in service-constrained inventory systems. Each develops an element of the two substitutions examined here — availability (inventory versus replenishment speed) and proximity (facilities versus last-mile travel). Each of these streams is well developed, and we draw on all four; we position the contribution by what it adds to each: the joint design of network density and replenishment speed under a carbon cap and a service requirement, together with the closed-form thresholds that relate the carbon-minimizing configuration to the cost-minimizing one.

2.1 Location–inventory design and risk pooling

A first stream establishes how demand risk pooling shapes distribution network design: jointly choosing facility location and safety stock, it characterizes the cost-optimal number and placement of distribution centers. Eppen (1979) characterizes the effect of centralization on expected cost in the multi-location newsvendor, establishing the square-root relationship between the number of stocking locations and pooled safety stock. Daskin et al. (2002) and Shen et al. (2003) embed working and safety stock into the facility-location decision, so that risk pooling shapes the optimal number and placement of distribution centers, with extensions to stochastic settings by Snyder et al. (2007), to capacitated warehouses by Özsen et al. (2008), and to continuous-approximation allocation by Tsao et al. (2012). Daganzo (2005) provides the continuous-approximation foundation in which last-mile distance scales with the inverse square root of the number of facilities. Closest to our spatial–inventory coupling, Straubert (2024) ties the number of stocking locations to an exact reorder-point/order-quantity inventory model through a delivery-lead-time penalty, solved by continuous approximation. The centralize-versus-decentralize question these models pose can also turn on forces beyond demand pooling: Schmitt et al. (2015) show analytically that supply-disruption risk diversification can favor decentralization for a risk-averse firm, in contrast to the classical pooling result. These works develop network density and safety-stock placement under risk pooling, with a cost or service objective and no carbon; they do not make replenishment speed a continuous decision coupled with carbon. The present study retains the risk-pooling structure and the density decision and adds an endogenous replenishment speed together with a carbon objective.

2.2 Carbon-aware network design

A second stream brings carbon into facility location and network design, characterizing how an emissions objective or cap can change the optimal network configuration. Cachon (2014) relates retail store density to greenhouse gas cost, with consumer travel to stores as the dominant emission channel and inventory entering through retail floor space. Turken et al. (2017) examine facility location and capacity under a carbon tax and emission limits, weighing centralization against decentralization through production scale economies; Harris et al. (2011) compare cost-optimal and emissions-aware network designs numerically; Wang et al. (2020) develop a carbon-compliant location–inventory model under demand and carbon-price uncertainty; Li et al. (2017) jointly optimize distribution-network design and two-echelon inventory under a CO_2 emission tax; and Hacardiaux and Tancrez (2020) assess the transport-emission savings of a shared, consolidated location–inventory network under horizontal cooperation, with Hacardiaux et al. (2022) balancing partner preferences over logistics cost and carbon footprint in the same cooperative setting. These studies make density or facility configuration the central decision and bring emissions into the objective, the constraints, or an ex post assessment; where transport is modeled as an operational choice, it enters through travel distances, discrete modes, or lead-time parameters rather than through a continuously optimized replenishment speed chosen jointly with density. The present study treats network density and replenishment speed as joint decisions and resolves the proximity substitution between facilities and last-mile travel analytically.

2.3 Inventory, transport mode, and emissions

A third stream couples transport-mode and lead-time choices with inventory control under emissions regulation, characterizing how carbon reshapes the trade-off between freight and the inventory it serves. Hoen et al. (2014) and Chen and Wang (2016) study transport mode selection alongside inventory under stochastic demand and emissions regulation; Bektaş and Laporte (2011) formalize the load–emission relationship in the pollution-routing problem, embedding fuel use and emissions as a function of vehicle load in the routing objective; Moradi et al. (2025) optimize driving speed and path selection under load- and grade-dependent fuel use at the routing level; Konur and Schaefer (2014) solve the economic order quantity model with less-than-truckload and truckload carriers under carbon cap, tax, and cap-and-trade regulation, and Schaefer and Konur (2015) extend this to a continuous-review, reorder-point inventory system under stochastic demand, characterizing the cost–emission Pareto frontier and analyzing lead-time and demand-variance sensitivity. Drent et al. (2023) analyze dynamic dual supply-mode selection under an emission target, and Hammami et al. (2015) bring carbon from manufacturing, ordering, and inventory into a multi-echelon production–inventory model with lead-time constraints. Across these papers, transport service is represented through route/load physics, carrier class, discrete supply mode, or lead-time parameters and constraints; replenishment speed is not a continuous decision chosen jointly with network density. The present study adds to this stream the continuous speed–density threshold that results when replenishment speed and network density are chosen jointly under a carbon cap and a service

requirement.

Boone and Ganeshan (2025) choose a discrete freight mode against inventory under an emission cap at a single location, with no network-density decision. The present study replaces the finite mode set with a continuous replenishment-speed decision, prices carbon on the holding as well as the transport side, embeds it in a network whose density is chosen jointly, and characterizes the resulting continuous speed–density threshold and complementarity.

2.4 Carbon policy, service, and safety-stock placement

A fourth stream prices carbon directly into inventory and service decisions, showing how a carbon price or cap shifts order quantities and safety-stock placement. Hua et al. (2011) and Bouchery et al. (2012) develop carbon-aware order-quantity models under cap-and-trade and multi-objective formulations, and Benjaafar et al. (2013) position cap, tax, and cap-and-trade as priced operational instruments in simple supply-chain models; Chen et al. (2013) derive the closed-form carbon-constrained economic order quantity, the single-location analog of the marginal carbon-intensity logic we extend to a network, and Arıkan and Jammernegg (2014) impose a product carbon-footprint constraint on a dual-sourcing newsvendor. Silbermayr et al. (2017) bring environmental constraints into inventory pooling with dependent demand. Within multi-echelon service models, Ni and Shu (2015) study the trade-off between service time and carbon in safety-stock placement under the guaranteed-service framework, and Wang et al. (2025) incorporate cap-and-trade into guaranteed-service placement. Probabilistic carbon limits have been modeled through chance constraints in lot sizing (Koca 2021) and in supply-chain network design (Shaw et al. 2016), typically inside large mixed-integer programs; we instead obtain a closed-form deterministic equivalent that scales the transport component by a single factor. A recent review (Sebatjane 2025) surveys carbon-aware inventory models and the regulatory instruments they treat; across this body of work, transport-emission modeling, service and inventory, network design, and demand or carbon uncertainty have developed in largely separate streams, and we bring these strands together. The service-placement models in particular price carbon against service and safety stock, with the joint design of network density and replenishment speed outside their scope.

Prior work covers carbon-aware density, mode and lead-time choice, service-constrained placement, and chance constraints on carbon. The present study couples endogenous replenishment speed with network density under carbon and service constraints and derives marginal carbon-intensity thresholds for the joint design. The resulting diagnostic identifies which activity supplies availability and proximity when carbon is priced: a speed threshold comparing the carbon intensity per dollar of holding with that of inbound transport, a pooling-adjusted density threshold comparing last-mile intensity with a weighted average of facility and holding intensities, and the complementarity that links them.

3 The Model

3.1 Operational environment

Region and markets. A firm serves N markets distributed over a service region of area A. Per-period demand at market i is a random variable with mean μ and standard deviation σ, identically distributed across markets with common pairwise correlation $\rho \in [0, 1)$. Demand is stationary and the setting is single-product. Markets and distribution centers are treated symmetrically, the standard simplification that admits closed-form continuous-approximation results (Daganzo 2005).

Distribution centers and geography. The firm opens n identical distribution centers (DCs), $1 \le n \le N$, each serving a sub-region of area A/n. By the continuous-approximation spatial law, the mean last-mile (DC-to-market) distance falls as the square root of the served area,

$$d(n) \;=\; \delta\, n^{-1/2}, \qquad \delta \propto \sqrt{A}, \tag{1}$$

so a denser network shortens delivery but only at a decreasing rate (Daganzo 2005).

Replenishment and the speed decision. Each DC is replenished directly from the source at a chosen speed $v \in [v_{\min}, v_{\max}]$. Individual DCs sit at different distances from the source; what enters cost and carbon is total inbound freight: with identical DCs each carrying throughput $N\mu/n$, this is $\sum_j (N\mu/n)\, D_j = N\mu\, \bar{D}$, the total throughput times the *mean* source-to-DC distance $\bar{D}$. Because the DCs fill a region of fixed size served from a fixed source, $\bar{D}$ is set by the region's geometry (order $\sqrt{A}$) and is approximately invariant to n; we write $D \equiv \bar{D}$ and treat it as constant.[1] The replenishment lead time is then

$$L(v) \;=\; D/v. \tag{2}$$

Faster replenishment is simultaneously more expensive and more carbon-intensive. The maintained assumption is only that the inbound cost rate $t(v)$ and emission rate $\varepsilon(v)$ (both per unit per mile) are increasing and convex in v; for the closed-form results, we use the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$ and $t(v) = t_0 + t_1 v^2$, the canonical form motivated by the cubic fuel-speed law of slow steaming (Psaraftis and Kontovas 2013; Cariou 2011). We read v as a general index of replenishment speed — vessel speed, or equivalently a faster and more carbon-intensive mode — so slow steaming is a motivating special case, not the only interpretation. Treating replenishment speed as a continuously

[1] Two modeling reductions are embedded here. First, the symmetric-DC reduction assigns every DC the representative lead time $L = \bar{D}/v$ in place of its own $L_j = D_j/v$. Safety stock enters through $\sqrt{L}$, which is concave in distance, so by Jensen's inequality $\sqrt{\bar{D}} \ge n^{-1} \sum_j \sqrt{D_j}$ (Jensen 1906): using the mean distance slightly *overstates* aggregate safety stock, by a gap that is second-order in the dispersion of the D_j and reduces as the distances equalize. Second, replenishment is point-to-point; were the DCs served on a consolidated route, the inbound distance would grow as $\sqrt{n}$ in the manner of the store-delivery tour of Cachon (2014), adding a term that reinforces centralization in Proposition 5 and leaving the fixed-density speed comparison of Proposition 3 unchanged; with inbound distance varying in n, however, the speed response would depend on density through $D(n)$, so the complementarity of Proposition 6 need not carry over unchanged.

optimized decision, chosen jointly with network density, couples the transport and facility-and-holding components in this distribution setting.

Inventory and service. Each DC operates a base-stock policy and must meet a cycle service level α — the in-stock probability at the end of a period, the Type-1 (cycle) service measure. With normally distributed demand the safety factor is $z = \Phi^{-1}(\alpha)$, with $\alpha > \frac{1}{2}$ so that $z > 0$, taken here as a service requirement (a parameter). A DC that pools $g = N/n$ markets faces aggregate per-period demand with standard deviation $\sigma\sqrt{g\,[\,1 + (g-1)\rho\,]}$, the statistical economy of scale in inventory (Eppen 1979). Covering lead-time demand, each DC holds safety stock $z\,\sigma\sqrt{L}\,\sqrt{g[1 + (g-1)\rho]}$. Summing over the n DCs and writing the *pooling expression*

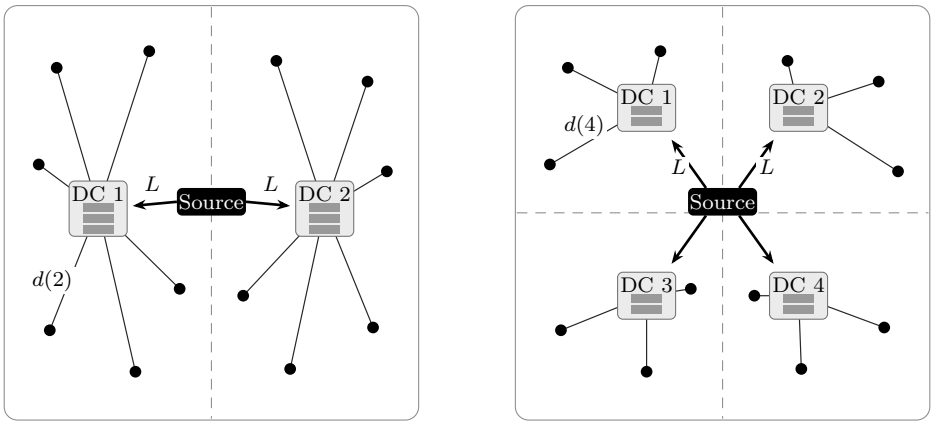
$$\Psi(n) \;=\; N\rho + n(1-\rho), \tag{3}$$

the system-wide safety stock is

$$\mathrm{SS}(n, v) \;=\; z\,\sigma\,\sqrt{D/v}\;\sqrt{N\,\Psi(n)}. \tag{4}$$

At $\rho = 0$, (4) reduces to $z\sigma\sqrt{L}\sqrt{nN}$, the square-root law: decentralizing from 1 to N DCs multiplies safety stock by $\sqrt{N}$. As $\rho \to 1$, $\Psi(n) \to N$ and safety stock becomes invariant to n — the pooling benefit vanishes.

Decisions. The firm chooses the network density $n \in [1, N]$ (relaxed to a continuous variable and rounded) and the replenishment speed $v \in [v_{\min}, v_{\max}]$. Service is a requirement, so α (hence z) is



$n = 2$: each DC serves 6 markets
more stock per DC, longer delivery legs

$n = 4$: each DC serves 3 markets
less stock per DC, shorter delivery legs

Faster replenishment (shorter L), at a given density:
system-wide safety stock shrinks, so holding cost and carbon fall, while freight cost and carbon rise.

More DCs (higher n), at a given speed:
delivery legs $d(n)$ shorten with territory area, so system-wide last-mile cost and carbon fall, while facility cost and carbon rise and weaker pooling raises system-wide holding cost and carbon.

Figure 1: Model representation for 12 markets (N) and 2 or 4 DCs ($n = 2$ or 4).

Table 1: Notation. The decision variables are the network density n and the replenishment speed v; all other quantities are exogenous parameters or derived.

Symbol	**Meaning**
Decision variables	
n	number of DCs (network density), $1 \le n \le N$
v	replenishment speed, $v \in [v_{\min}, v_{\max}]$
Demand (exogenous)	
N	number of markets
$\mu,\ \sigma$	mean, standard deviation of demand per market per period
ρ	cross-market demand correlation, $\rho \in [0, 1)$
Geography, cost, carbon (exogenous)	
$A,\ D,\ \delta$	region area; inbound distance; last-mile distance constant
$F,\ \varphi$	fixed cost, carbon per DC per period
$h,\ \eta$	holding cost, carbon per unit per period
$t(v),\ \varepsilon(v)$	inbound transport cost, carbon per unit-mile (convex in v)
$t_L,\ \varepsilon_L$	last-mile transport cost, carbon per unit-mile
α	cycle service level (in-stock prob.); $z = \Phi^{-1}(\alpha)$
$\tau,\ B$	carbon price; carbon cap
Derived	
$g = N/n$	markets served per DC
$L = D/v$	replenishment lead time
$d(n) = \delta\, n^{-1/2}$	mean last-mile distance
$\Psi(n) = N\rho + n(1-\rho)$	pooling expression
$\mathrm{SS}(n, v)$	system-wide safety stock, eq. (4)

a parameter in our model. Table 1 collects the notation.

3.2 Cost and carbon

Both the cost and carbon functions are system-wide flows over a single representative period, consistent with the stationary single-product environment above: F, φ are charged per DC per period, h, η per unit held per period, and the transport terms accrue against the per-period throughput. The representative distances in Figure 1 therefore enter global cost and carbon through aggregate freight tasks, not through a per-territory objective. Throughput itself is conserved at $N\mu$ regardless of the configuration: the firm moves the mean demand of all N markets through the network, whatever density n and speed v it chooses. With per-period throughput $N\mu$,

$$K(n, v) = \underbrace{F\, n}_{\text{fixed}} + \underbrace{h\, \mathrm{SS}(n, v)}_{\text{holding}} + \underbrace{t(v)\, N\mu D}_{\text{inbound}} + \underbrace{t_L\, N\mu\, d(n)}_{\text{last-mile}}, \tag{5}$$

$$C(n, v) = \underbrace{\varphi\, n}_{\text{fixed}} + \underbrace{\eta\, \mathrm{SS}(n, v)}_{\text{holding}} + \underbrace{\varepsilon(v)\, N\mu D}_{\text{inbound}} + \underbrace{\varepsilon_L\, N\mu\, d(n)}_{\text{last-mile}}, \tag{6}$$

where F, φ are fixed cost/carbon per DC, h, η are holding cost/carbon per unit per period, and t_L, ε_L are last-mile cost/carbon rates (Shen et al. 2003; Hua et al. 2011; Benjaafar et al. 2013).

Each term multiplies a rate by the activity it prices; the two functions differ only in whether the rate is measured in dollars or kilograms.

Facilities. The facility terms Fn and φn are the periodic cost and carbon of operating the open DCs. They rise linearly in n and do not depend on v. Facility carbon φ is distinct from holding carbon η: φ captures baseline building energy and embodied carbon, whereas η prices the incremental carbon of storing a unit (for example refrigeration or handling).

Holding. The holding terms $h\mathrm{SS}(n, v)$ and $\eta\mathrm{SS}(n, v)$ value the system-wide safety stock of (4). They rise in n through the loss of pooling in $\Psi(n)$ and fall in v through the lead-time factor $\sqrt{D/v}$. This is the channel through which replenishment speed couples transport and inventory.

Inbound transport. The inbound terms multiply the representative freight task $N\mu D$ by the speed-dependent rates $t(v)$ and $\varepsilon(v)$. Since D is treated as invariant to n, inbound transport does not vary with network density; it rises in v through the increasing convex rates $t(v)$ and $\varepsilon(v)$, opposing the holding term that falls in v.

Last-mile transport. The last-mile terms multiply throughput $N\mu$ by the mean DC-to-market distance $d(n) = \delta n^{-1/2}$ and the rates t_L, ε_L. A denser network shortens this leg at the continuous-approximation rate, so the last-mile term is decreasing and convex in n and independent of v under constant per-mile last-mile rates.

Decomposition and scope. The carbon function splits into a transport component $\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)$ and a facility-and-holding component $\varphi n + \eta\mathrm{SS}(n, v)$; the cost function has the analogous partition. Slowing replenishment reduces inbound transport carbon but raises safety stock, while dispersion reduces last-mile distance but raises facility count and pooling loss. Three scope points follow. Cycle stock under a fixed review period is approximately $N\mu/2$ system-wide, or $(N\mu/n)/2$ per DC, and is invariant to both decisions. Pipeline inventory $N\mu D/v$ is omitted; when valued at the same dollar and carbon holding rates as stored inventory, it changes the level of v^* but not the speed threshold (Section 4). Production and procurement are also excluded because throughput is fixed at $N\mu$ and therefore constant in both decisions.

3.3 The program

The firm minimizes cost subject to a carbon cap B and the service requirement,

$$\min_{1 \le n \le N,\ v_{\min} \le v \le v_{\max}} K(n, v) \quad \text{s.t.} \quad C(n, v) \le B. \tag{7}$$

Here K and C are expected flows over the model's accounting period: under the stationary demand assumption they are the long-run average cost and carbon of a configuration, and they can be scaled to an annual expected-carbon cap by multiplying by the number of periods. Because demand is

random, realized carbon over a specified accounting horizon fluctuates about its expectation; we develop the analysis for the expected-value cap and return below to a chance-constrained cap that bounds realized carbon with specified confidence (Proposition 10).

Equivalently, pricing carbon at $\tau \geq 0$ gives the priced objective $J = K + \tau C$. At an interior optimum, the cap multiplier equals the price ($\lambda = \tau$), as in carbon-constrained inventory models (Chen et al. 2013); sweeping τ traces the cost–carbon frontier, along which the optimal configuration moves continuously (Proposition 2). Because carbon at the priced optimum then falls continuously from its cost-minimizing level to its minimum, every cap in that range is met by some price, and the capped and priced problems share the same unique solution; jumps arise only from the integer number of distribution centers. Define the blended rates

$$F_\tau = F + \tau\varphi, \qquad H_\tau = h + \tau\eta, \qquad T_\tau(v) = t(v) + \tau\varepsilon(v), \qquad \ell_\tau = t_L + \tau\varepsilon_L.$$

Then

$$J(n, v; \tau) \;=\; F_\tau n + H_\tau \,\mathrm{SS}(n, v) + T_\tau(v) N\mu D + \ell_\tau N\mu d(n). \tag{8}$$

Setting $\tau = 0$ recovers cost; the limit $\tau \to \infty$ recovers carbon up to scale.

4 Analysis

We now characterize how carbon pricing changes the joint design. We first show that the priced problem has a single best design at every carbon price (Proposition 2). The first-order conditions then decompose into two marginal substitutions, each analyzed with the other decision held fixed: Proposition 3 gives the speed threshold on the availability margin, Proposition 4 records the service and volatility comparative static, and Proposition 5 gives the pooling-adjusted density threshold on the proximity margin. Proposition 6 establishes the complementarity through which the two margins interact, and Proposition 7 combines them into an exact classification of the carbon-minimizing direction of both decisions. Corollary 8 gives the magnitudes in closed form when pooling is absent, and Proposition 9 describes how the design moves as the carbon price rises. Section 5 applies them to a calibrated network. Proofs are in Appendix A.

Lemma 1 (Carbon intensity per dollar)**.** *Pricing carbon at τ replaces each cost rate in* (5) *by the blended rate* $\mathrm{rate}_\$ + \tau\,\mathrm{rate}_{\mathrm{kg}}$*, yielding the objective J of* (8)*. In the interior comparisons below, each holding the other decision fixed, the* direction *in which the carbon-minimizing choice departs from the cost-minimizing one is governed by four carbon-intensity-per-dollar ratios — holding η/h, inbound transport ε_0/t_1, last-mile ε_L/t_L, and facilities φ/F — while the magnitude and the realized joint optimum also depend on pooling, the bounds, and the complementarity of the two decisions.*

The carbon-minimizing choice on each decision therefore depends on each activity's carbon intensity *relative to its cost*, not on its absolute rates.

Writing $G(n,v) = \partial \mathrm{SS}(n,v)/\partial n$, differentiation gives the stationarity conditions

$$F_\tau + H_\tau G(n,v) = \frac{\ell_\tau N \mu \delta}{2}\, n^{-3/2}, \tag{9}$$

$$N\mu D\, T'_\tau(v) = \frac{H_\tau\, \mathrm{SS}(n,v)}{2v}, \tag{10}$$

the first balancing the marginal facility and lost-pooling costs against the marginal last-mile saving, and the second balancing the marginal transport cost of speed against the marginal holding saving it obtains. The joint first-order conditions thus reveal two marginal substitutions. The speed condition (10) governs the *availability margin*: safety stock and replenishment speed are alternative ways to provide service. The density condition (9) governs the *proximity margin*: facilities and last-mile distance are alternative ways to reach markets. The thresholds below compare the carbon intensity per dollar of the activities on each margin, and complementarity determines how the two margins interact in the joint configuration.

The stationarity conditions identify the optimum because the priced problem has a single best design at every carbon price, although J is not jointly convex over the whole feasible region. Let $J^{\mathrm{r}}(n;\tau) = \min_{v_{\min}\le v\le v_{\max}} J(n,v;\tau)$ be the *reduced objective*, the priced cost of operating n distribution centers when speed is chosen optimally for that network. A function on an interval is *strictly quasiconvex* if it strictly decreases and then strictly increases, either phase possibly empty, so that it has a single minimizer.

Proposition 2 (A single best design)**.** *Let t and ε be increasing, differentiable, and convex, let all cost and carbon rates be positive, and let $v_{\min} > 0$. For every carbon price $\tau \ge 0$, and for the carbon objective C:* (i) *for each n, the objective is strictly convex in v, and the speed response $v^*(n)$ is unique, continuous, and nondecreasing in n;* (ii) *the objective at any fixed speed, and the reduced objective, are strictly quasiconvex in n on $[1,N]$;* (iii) *the problem has a unique minimizer, and for the priced problem the minimizer $\big(n^*(\tau), v^*(\tau)\big)$ is continuous in τ.*

Proposition 2 has a direct reading. Adding a distribution center shortens every delivery but adds a facility charge and the safety stock lost to pooling. Scaled by $n^{3/2}$, these marginal costs rise with the size of the network while the marginal last-mile saving does not, so once an additional center stops paying for itself, no larger network makes it pay again. Three consequences follow. The stationarity conditions (9)–(10) identify the optimum whenever it is interior; the comparative statics below are global rather than local to a particular optimum; and in the continuous model the design moves without jumps as the carbon price rises, so discontinuities arise only from the integer number of distribution centers. Because the reduced objective is flat near its minimum, treating that number as continuous costs little: rounding it to the best integer raises cost or carbon by at most 0.03% in the calibrated network of Section 5, in line with the accuracy reported for continuous-approximation designs (Ouyang and Daganzo 2006; Ansari et al. 2018).

Availability: inventory or speed. The speed condition isolates the availability margin. Increasing speed raises inbound transport cost and carbon but lowers the safety stock required for service; the speed threshold identifies which activity is more carbon-intensive per dollar on this margin.

Proposition 3 (Speed reversal)**.** *Fix the density n and write the speed-dependent cost and carbon as*

$$\hat{K}(v) = h\,\kappa(n)\,v^{-1/2} + N\mu D\,t(v), \qquad \hat{C}(v) = \eta\,\kappa(n)\,v^{-1/2} + N\mu D\,\varepsilon(v),$$

with $\kappa(n) = z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}$. If t and ε are increasing, differentiable, and convex, then at a common density and for interior optima the carbon-minimizing speed exceeds the cost-minimizing speed if and only if holding is more carbon-intensive per dollar than inbound transport at the margin*,*

$$\frac{\eta}{h} \;>\; \frac{\varepsilon'(v_{\mathrm{K}})}{t'(v_{\mathrm{K}})}, \tag{11}$$

evaluated at the cost-minimizing speed v_{K}. With the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$, $t(v) = t_0 + t_1 v^2$, the threshold reduces to $\eta/h > \varepsilon_0/t_1$ and the unconstrained optimum is closed form,

$$v^*(n) \;=\; \left[\frac{(h+\tau\eta)\,z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}}{4\,(t_1+\tau\varepsilon_0)\,N\mu D}\right]^{2/5}. \tag{12}$$

On $[v_{\min}, v_{\max}]$ the optima are the projections of the unconstrained speeds onto the interval; under the threshold (11) *the comparison $v_C \geq v_K$ then holds weakly, and strictly when neither projection binds.*

Because (12) depends on the carbon price only through the blended ratio $(h+\tau\eta)/(t_1+\tau\varepsilon_0)$, which is monotone in τ, the fixed-density speed optimum moves monotonically between the cost and carbon endpoints as the carbon price rises: the endpoint comparison of the proposition also orders every finite price.

Total carbon is U-shaped in speed. When holding is more carbon-intensive per dollar than the freight that would replace it, a tighter carbon target can select *faster* replenishment; Section 5 shows that this occurs mainly when speed is bought with labor rather than fuel. Volatility and the service target raise the level of speed at both optima but not its direction. The comparison is at common density; in the joint problem, the density channel reinforces the speed reversal when carbon also disperses the network and attenuates it otherwise, and Proposition 7 separates the two effects exactly.

Service and volatility enter through the same channel. Write $J^{\mathrm{r}}(n;s) \equiv J\big(n, v^*(n;s); s\big)$, with $s = z\sigma$, for the reduced objective obtained after substituting the speed response into the priced objective.

Proposition 4 (Service and volatility as carbon parameters)**.** *Under the quadratic instantiation, $v^*(n)$ is increasing in the safety factor $z = \Phi^{-1}(\alpha)$ and in the demand standard deviation σ wherever*

it is interior, and at every n both raise the holding term in (9)*. Consequently, at every carbon price, a higher z or σ raises the inventory penalty of a denser network and lowers the optimal density, strictly when it is interior; in particular, it shifts the carbon-minimizing density toward consolidation.*

A service requirement therefore affects the carbon-minimizing design as well as inventory cost; endogenous service choice is a natural extension (Section 7).

Proximity: facilities or last-mile travel. The density condition isolates the proximity margin. Increasing density shortens the last mile but raises facility cost and carbon and weakens pooling; the density threshold identifies when carbon places more weight on last-mile distance than on the facility-and-pooling cost of proximity, and pooling keeps the comparison from reducing to a single pair of rates.

Proposition 5 (Density and dispersion)**.** *Fix the speed v and let*

$$G(n,v) \;=\; \frac{\partial\,\mathrm{SS}(n,v)}{\partial n} \;=\; z\sigma\sqrt{D/v}\,\sqrt{N}\,\frac{1-\rho}{2\sqrt{\Psi(n)}} \;\geq\; 0 \tag{13}$$

be the marginal safety-stock penalty of an additional DC. The minimizer of $J(\cdot,v)$ is unique; it is interior if and only if $J_n(1,v) < 0 < J_n(N,v)$, and otherwise lies at a boundary, $n^* \in \{1,N\}$. For an interior cost-minimizing density $n_K(v)$, the carbon-minimizing density exceeds it, $n_C(v) > n_K(v)$, if and only if last-mile transport is more carbon-intensive per dollar than the pooling-adjusted facility threshold*,*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\,G\big(n_K(v),v\big)}{F + h\,G\big(n_K(v),v\big)}. \tag{14}$$

The right-hand side is a weighted average of the facility ratio φ/F and the holding ratio η/h, with weight on η/h rising in the pooling penalty G: it equals φ/F when pooling is absent and tilts monotonically toward η/h as G grows (lower correlation, higher safety factor or volatility, or slower replenishment). In the no-pooling limit ($\rho \to 1$, so $G \to 0$) the optimum is closed form when interior, $n^* = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$ (with the blended last-mile and facility rates $\ell_\tau = t_L + \tau\varepsilon_L$ and $F_\tau = F + \tau\varphi$ of (8)*), and* (14) *reduces to*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi}{F}. \tag{15}$$

The density decision is the centralize-versus-decentralize question in carbon terms. By (14), the carbon-minimizing network is more dispersed when the last-mile ratio ε_L/t_L exceeds a facility threshold that pooling shifts from φ/F toward η/h. Pooling therefore enters as an explicit weight on the holding ratio, not as a uniform centralizing force: when holding is more carbon-intensive per dollar than facilities, stronger pooling raises the threshold for dispersion; otherwise it lowers it. A corner optimum, when it arises, is fully centralized or fully decentralized.

Complementarity of speed and density. The two margins interact through safety stock: faster replenishment reduces the inventory penalty of density, and higher density increases the inventory exposure affected by speed. Proposition 6 makes the interaction precise.

Proposition 6 (Complementarity)**.** *For $\rho < 1$, the speed response $v^*(n)$ is nondecreasing in n and the density response $n^*(v)$ is nondecreasing in v, each strictly where it is interior.*

Faster replenishment shrinks safety stock and lowers the inventory penalty of dispersion; a denser network raises the inventory at stake and raises the value of speed. Speed investments and network density are therefore best evaluated jointly.

The joint design. The marginal thresholds hold one decision fixed. In the joint design both decisions adjust, and complementarity links them: a carbon price that disperses the network also raises the value of speed, and one that speeds replenishment lowers the inventory penalty of dispersion. Proposition 7 resolves the joint comparison exactly. Write $r_h = \eta/h$, $r_t = \varepsilon_0/t_1$, $r_L = \varepsilon_L/t_L$, and $r_F = \varphi/F$ for the four ratios of Lemma 1, so that the two diagnostic ratios are

$$R_{\text{speed}} = \frac{r_h}{r_t}, \qquad R_{\text{density}} = \frac{r_L}{r_F},$$

and let (n_K, v_K) and (n_C, v_C) denote the cost- and carbon-minimizing configurations. Two quantities of the cost-minimizing network enter. The first is the *pooling weight*

$$\omega = \frac{h\,G(n_K, v_K)}{F + h\,G(n_K, v_K)} = 1 - \Big(\frac{n_K}{n_K^0}\Big)^{3/2}, \qquad n_K^0 = \Big(\frac{t_L N \mu \delta}{2F}\Big)^{2/3}, \tag{16}$$

the share of the marginal cost of a distribution center that is lost pooling rather than facility cost. Since n_K^0 is the unconstrained density the firm would choose if pooling were absent, ω measures how far pooling holds the network below that size. The second, for interior n_K, is the pair of *speed bands*

$$\underline{R} = \sqrt{\Psi(n_K)/N} \;<\; 1 \;<\; \overline{R} = \sqrt{\Psi(n_K)/\Psi(1)}, \tag{17}$$

which bound how far a change in density can move the carbon-minimizing speed.

Proposition 7 (Joint carbon-minimizing design)**.** *Under the quadratic instantiation, suppose that $\rho < 1$, that the cost-minimizing density n_K is interior, and that the speed responses of the cost and the carbon objective are interior at n_K and at n_C.*

(i) Density. *The carbon-minimizing network is more dispersed than the cost-minimizing one, $n_C > n_K$, if and only if*

$$r_L \;>\; (1-\omega)\, r_F \;+\; \omega\, r_h^{4/5}\, r_t^{1/5}, \tag{18}$$

with $n_C = n_K$ at equality and $n_C < n_K$ when the inequality is reversed.

(ii) Speed. *The carbon-minimizing speed satisfies*

$$\ln \frac{v_C}{v_K} = \frac{2}{5} \ln R_{\text{speed}} + \frac{1}{5} \ln \frac{\Psi(n_C)}{\Psi(n_K)}, \tag{19}$$

so that $v_C > v_K$ *if and only if* $R_{\text{speed}} > \sqrt{\Psi(n_K)/\Psi(n_C)}$.

(iii) Classification. *If* (18) *holds and* $R_{\text{speed}} \geq 1$*, the carbon-minimizing design is faster and more dispersed; if* (18) *is reversed and* $R_{\text{speed}} \leq 1$*, it is slower and more consolidated. Whatever the density, it is faster if* $R_{\text{speed}} > \overline{R}$ *and slower if* $R_{\text{speed}} < \underline{R}$*. Only when the two conditions point in opposite directions and* $\underline{R} \leq R_{\text{speed}} \leq \overline{R}$ *does the direction of speed depend on the size of the density change, which* (19) *then determines. At equality in* (18)*, the direction of speed is that of* R_{speed} *relative to one.*

Proposition 7 answers, for a given product, the two questions a firm asks of its network. The density condition (18) is Proposition 5 with one change. Adding a distribution center incurs a facility charge and lost pooling, in the proportions $1-\omega$ and ω of its marginal cost. Along the optimal speed response, the firm meets part of the lost pooling by replenishing slightly faster: an additional center raises holding by four-fifths of the cost of lost pooling and inbound freight by one-fifth. The carbon-minimizing firm makes the same split at its own speed, which differs from the cost-minimizing speed by the factor $R_{\text{speed}}^{2/5}$, so per dollar of lost pooling its carbon is the weighted geometric mean $r_h^{4/5} r_t^{1/5}$. The carbon-minimizing network is more dispersed when a dollar of last-mile delivery emits more than this blend of facility and lost-pooling carbon. One consequence has no counterpart in the marginal analysis: the carbon intensity of inbound freight bears on how many distribution centers to operate, because a lower-emission line haul makes speed a less carbon-intensive way to recover lost pooling and so lowers the threshold for dispersion. The exponents are these cost shares: with freight cost and carbon proportional to v^p rather than v^2, holding's share, and with it the exponent on r_h, becomes $2p/(2p+1)$, and inbound freight's becomes $1/(2p+1)$.

The speed decomposition (19) separates the two reasons a carbon-minimizing chain replenishes at a different speed. The intensity effect, $\frac{2}{5}\ln R_{\text{speed}}$, is the speed reversal of Proposition 3. The density effect, $\frac{1}{5}\ln[\Psi(n_C)/\Psi(n_K)]$, is complementarity: a more dispersed network holds more safety stock and so values speed more. The bands (17) bound the density effect, because no network pools more than full consolidation or less than full dispersion. In the calibrated network of Section 5, $\underline{R} = 0.61$ and $\overline{R} = 1.08$: a product whose holding is more than about 8% more carbon-intensive per dollar than the speed it would buy is replenished faster under carbon whatever happens to its network, and one whose ratio falls below 0.61 is replenished more slowly. Speed bought with fuel places every product calibrated there far below the lower band, and speed bought with labor places most above the upper one.

Corollary 8 (Magnitudes without pooling)**.** *In the no-pooling limit (*$\rho \to 1$*),* $\omega = 0$*, the two decisions separate, and for interior optima*

$$\frac{n_C}{n_K} = R_{\text{density}}^{2/3}, \qquad \frac{v_C}{v_K} = R_{\text{speed}}^{2/5}.$$

Carbon separates into a proximity component $P(n) = \varphi n + \varepsilon_L N \mu \delta n^{-1/2}$ *and an availability component* $A(v) = \eta z \sigma N \sqrt{D/v} + \varepsilon_0 N \mu D\, v^2$*, and the carbon of the cost-minimizing design relative to*

the carbon minimum $C_{\min}$ *is*

$$\frac{C(n_K, v_K)}{C_{\min}} = \theta\,\frac{x + 2x^{-1/2}}{3} + (1-\theta)\,\frac{4y^{-1/2} + y^2}{5}, \qquad x = R_{\text{density}}^{-2/3}, \quad y = R_{\text{speed}}^{-2/5}, \tag{20}$$

where $\theta = P(n_C)/C_{\min}$ *is the proximity share of the minimum. To second order in the logarithms of the ratios, the excess carbon of the cost-minimizing design is* $\theta\,(\ln R_{\text{density}})^2/9 + (1-\theta)\,\frac{2}{25}(\ln R_{\text{speed}})^2$.

Corollary 8 gives the size of the redesign. Absent pooling and with both optima interior, a product whose last mile is four times as carbon-intensive per dollar as its facilities runs $4^{2/3} \approx 2.5$ times as many distribution centers when minimizing carbon. To second order, the excess carbon of the cost-minimizing design grows with the square of the logarithm of each ratio: it is small near parity and rises quickly away from it, so that the cost-minimizing network emits about 5% more proximity carbon than the minimum when $R_{\text{density}} = 2$ and about 19% more when $R_{\text{density}} = 4$. Carbon-aware redesign therefore matters most for products whose ratios lie far from one. With pooling the two decisions interact and these expressions are no longer exact; Section 5 illustrates the magnitudes in that case.

The carbon-price path. A firm facing a rising carbon price needs to know how the design moves along the way, not only where it ends. For an activity with carbon intensity per dollar r_x, write $s_x(\tau) = \tau r_x/(1+\tau r_x)$ for the *carbon share* of its blended rate — for holding, $s_h = \tau\eta/(h+\tau\eta)$ — which rises from zero to one as the price rises.

Proposition 9 (Carbon-price path)**.** *Under the quadratic instantiation:* (i) *at a fixed density and an interior speed response,* $\mathrm{d}\ln v^*/\mathrm{d}\ln\tau = \frac{2}{5}\,(s_h - s_t)$*;* (ii) *in the no-pooling limit and for an interior density,* $\mathrm{d}\ln n^*/\mathrm{d}\ln\tau = \frac{2}{3}\,(s_L - s_F)$*;* (iii) *with pooling and interior speeds, if* r_L *exceeds each of* r_F*,* r_h*, and* r_t*, the optimal density* $n^*(\tau)$ *is nondecreasing along the whole price path, and if* r_L *falls short of each, it is nonincreasing.*

The carbon share of a blended rate measures how much of that activity's priced cost is carbon. By part (i), at a given network, a rising carbon price speeds replenishment exactly when carbon is a larger share of the priced cost of holding than of inbound freight, and the response is gradual: at a low price both shares are near zero and speed barely moves, while at a high price both approach one and speed settles at its carbon-minimizing level. Part (iii) gives a condition a firm can check once. When its last mile is the most carbon-intensive of the four activities per dollar, no increase in the carbon price calls for consolidation, and where the density is interior every increase disperses the network; when the last mile is the least carbon-intensive, the reverse holds. Outside these two cases, part (iii) does not fix the direction along the whole path.

Demand uncertainty: a chance-constrained cap. The cost and carbon functions are expected per-period flows, so the cap in (7) bounds *expected* carbon. Because demand is random, realized carbon fluctuates about $C(n, v)$: the facility charge and the base-stock safety-stock target

are planned and treated as deterministic, but inbound and last-mile freight scale with realized throughput $Q = \sum_i X_i$, the sum of the realized market demands, which has mean $N\mu$ and variance $\sigma_Q^2 = \sigma^2 N\,[\,1 + (N-1)\rho\,]$. Realized carbon is then

$$\tilde{C}(n, v) \;=\; \varphi n + \eta\,\mathrm{SS}(n, v) + \big[\varepsilon(v)D + \varepsilon_L d(n)\big]\,Q, \tag{21}$$

and a firm that requires realized carbon to stay within B with confidence $1 - \beta_c$ — where β_c is the carbon-side tail probability, written with the subscript c to distinguish it from the inventory fill-rate β of the lot-sizing literature — imposes the chance constraint $\Pr\{\tilde{C}(n, v) \le B\} \ge 1 - \beta_c$, the carbon-side counterpart of the cycle service level α on the inventory side.

Proposition 10 (Deterministic equivalent of the chance constraint)**.** *This chance constraint is equivalent to*

$$\varphi n + \eta\,\mathrm{SS}(n, v) + m\big[\varepsilon(v)N\mu D + \varepsilon_L N\mu\, d(n)\big] \;\le\; B, \tag{22}$$

the carbon function (6) *with the transport component scaled by* $m = q_{1-\beta_c}/(N\mu)$*, where* $q_{1-\beta_c} = \inf\{q : \Pr(Q \le q) \ge 1 - \beta_c\}$ *is the lower* $(1 - \beta_c)$ *quantile of realized throughput* Q*, and the facility and holding terms unchanged. Under the normal approximation* $Q \sim \mathcal{N}(N\mu, \sigma_Q^2)$ *this is* $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$*, with* $z_{\beta_c} = \Phi^{-1}(1 - \beta_c)$ *and* $\mathrm{CV}_Q = \sigma_Q/(N\mu)$*; then* $m \ge 1$ *whenever* $\beta_c \le \frac{1}{2}$*.*

The chance constraint therefore leaves the structure intact: (22) is the four-component carbon function with the inbound and last-mile carbon rates scaled, $\varepsilon_0 \mapsto m\varepsilon_0$ and $\varepsilon_L \mapsto m\varepsilon_L$, while cost and the facility and holding terms are unchanged. Because Q is realized market throughput, exogenous to the firm's choice of n and v, the factor m is a decision-invariant scalar on the transport carbon rates; the priced objective is again (8) under this substitution, so Propositions 2–9 carry over with ε_0 and ε_L replaced by $m\varepsilon_0$ and $m\varepsilon_L$, and only the thresholds shift.

Corollary 11 (Uncertainty tightens the speed threshold and relaxes the density threshold)**.** *Under the chance-constrained cap the speed threshold* (11) *becomes* $\eta/h > m\,\varepsilon'(v_{\mathrm{K}})/t'(v_{\mathrm{K}})$ *(*$m\,\varepsilon_0/t_1$ *under the quadratic), and the pooling-adjusted density threshold* (14) *becomes*

$$m\,\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\,G\big(n_K(v), v\big)}{F + h\,G\big(n_K(v), v\big)}, \tag{23}$$

which reduces to $m\,\varepsilon_L/t_L > \varphi/F$ *in the no-pooling limit (*$\rho \to 1$*); in the joint design the speed ratio becomes* R_{speed}/m *and the density condition* (18) *becomes* $m\,r_L > (1-\omega)\,r_F + \omega\,m^{1/5} r_h^{4/5} r_t^{1/5}$*. Under the normal approximation,* $m \ge 1$ *for* $\beta_c \le \frac{1}{2}$*, with equality at* $\beta_c = \frac{1}{2}$*, so the chance constraint tightens the speed-reversal threshold and relaxes the dispersion threshold relative to the expected-value cap, and the effect grows as* m *rises with the demand coefficient of variation* σ/μ*, the correlation* ρ*, and the confidence* $1 - \beta_c$*. In the joint design the carbon-minimizing speed itself falls as* m *rises.*

Demand uncertainty therefore affects carbon through two channels. It enters the inventory-holding component through the safety stock $z\sigma\sqrt{L}$, the channel behind the speed reversal, and, under a

chance-constrained cap, it enters the transport component through the margin $z_{\beta_c}\,\mathrm{CV}_Q$ that the firm holds against realized emissions just as it holds safety stock against demand. More variable or more correlated demand raises this margin and, like a higher service requirement, shifts the carbon-minimizing configuration; the expected-value cap is the limiting case $m = 1$. Carrying the realized variability of on-hand inventory as well would place an analogous margin on the inventory-holding component; we keep the safety stock at its planned level, consistent with the base-stock policy. The deterministic equivalent itself is distribution-free in the throughput quantile $q_{1-\beta_c}$; the normal approximation enters only through the closed form $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$, which is most accurate for moderate N and modest CV_Q and which a firm with strongly skewed demand can replace by the empirical throughput quantile without altering the structure.

The cap applies over the same accounting period as the flow model — here a week. For a cap on cumulative carbon over T periods with demands independent across periods, aggregate throughput has coefficient of variation $\mathrm{CV}_Q/\sqrt{T}$, so the multiplier tightens toward one as the horizon lengthens: in the calibrated network of Section 5, the 95% multiplier falls from $m = 1.37$ for a weekly cap to $m = 1.05$ for an annual cap ($T = 52$). The chance-constrained results are therefore most relevant to high-frequency operational carbon budgets, shipment-cycle accounting, and corporate carbon controls, and less binding for annual compliance accounting.

In-transit inventory. The holding term is modeled through safety stock; in-transit (pipeline) inventory, $N\mu L = N\mu D/v$, falls in v and is omitted from our model. If pipeline inventory were carried at the warehouse holding rates, it would add $(h+\tau\eta)\,N\mu D/v$ to the holding term; the speed first-order condition then acquires a v^{-2} term, and stationarity again equates $(h + \tau\eta)/(t_1 + \tau\varepsilon_0)$ to a quantity strictly increasing in v — the same ratio that governs (11). Valued at common rates, pipeline inventory therefore shifts the level of v^* (the closed form (12) is replaced by a numerical root) but preserves the carbon-intensity threshold; carrying capital cost with negligible warehouse carbon instead raises only the cost-minimizing speed, shifting the threshold against the speed reversal. We treat pipeline inventory as a level effect under the equal-rate convention and leave alternative carbon accounting for in-transit stock as an extension.

Load-factor sensitivity. Our model treats the per-unit last-mile rates t_L and ε_L as independent of shipment size. Last-mile cost and carbon, however, accrue largely per *vehicle*-mile — driver time and fuel — so the per-unit rates scale with vehicle-miles per unit, which rise as dispersion thins each territory's dispatch volume and average vehicle fill falls; load-sensitive rates therefore introduce a consolidation force outside the constant-rate model. Because a fill multiplier depends on n, it enters the marginal density condition through both the level and the derivative of the last-mile term. When a common fill multiplier is applied to the cost and the carbon rate alike, both optima shift toward consolidation in the reported computation, while their relative direction remains consistent with the carbon-intensity comparison. Per vehicle-mile, cost is dominated by driver time while fuel use falls somewhat in an emptier vehicle — the basis of tonne-kilometre emission accounting (Smart Freight Centre 2022) — so when fill deteriorates, the per-unit cost rate typically inflates

at least as strongly as the per-unit carbon rate. A multiplier falling more heavily on carbon is therefore a stress case for the dispersion result; under it, the density comparison can shift toward consolidation. A full treatment would make shipment size, routing, and consolidation additional decisions, which we leave as an extension.

5 A Calibrated Application

This section applies the results to distribution activities measured from public US data. Section 5.1 estimates the carbon intensity per dollar of each activity, Section 5.2 classifies representative products by Proposition 7, and Section 5.3 computes cost- and carbon-minimizing designs for a calibrated national replenishment network in physical units. The directions in Section 5.2 follow from the propositions once the ratios are known; the magnitudes in Section 5.3 describe the network we calibrate and illustrate the size of the effects in that network.

5.1 Carbon intensity per dollar of distribution activities

Each ratio is the carbon an activity emits per dollar it costs. We compute it from the bottom up, as the activity's energy use times the emission factor of that energy, divided by the activity's cost, and compare the result with the industry factors of the US environmentally extended input–output model (U.S. Environmental Protection Agency 2024a). One observation organizes the results. A dollar spent on energy carries about 2.8 kg CO_2e as diesel, 2.9–4.6 kg as grid electricity, and 5.0–5.8 kg as marine fuel, while the labor and capital that make up the rest of logistics cost carry little carbon of their own. An activity's carbon intensity per dollar is therefore close to its energy share of cost times a few kilograms. Table 2 reports the ratios.[2]

Holding is the least carbon-intensive activity per dollar, because its cost is mostly the capital tied up in stock and the labor and space to store it. The ratio falls with value density: a pallet of pharmaceuticals carries refrigeration carbon similar to a pallet of dairy but many times its holding cost. Buying speed with fuel is the most carbon-intensive, because the added cost of a faster vessel, of truck instead of intermodal rail, or of air instead of truck is mostly fuel. Buying speed with labor

[2]Emission factors: diesel 10.21 kg CO_2 per gallon and the US grid 0.371 kg CO_2e per kWh (U.S. Environmental Protection Agency 2022); marine fuel 3.114 t CO_2 per tonne (European Commission 2016). Prices: diesel \$3.66 per gallon (2025 average) (U.S. Energy Information Administration 2026), electricity 8.1–12.8 cents per kWh (U.S. Energy Information Administration 2025), and marine fuel \$535–626 per tonne (Ship & Bunker 2025). Holding: storage energy of about 30 kg CO_2e per pallet-year in ambient and 109 kg in refrigerated warehouses, from national energy-use medians (U.S. Environmental Protection Agency 2024b) and an operator's reported emissions and pallet positions (Lineage, Inc. 2024); storage at \$240 (ambient) and \$480 (refrigerated) per pallet-year; and capital at 10% of pallet value. Line haul: truckload at \$2.26 per mile, of which fuel is \$0.48 (American Transportation Research Institute 2025); intermodal 10–30% cheaper than truckload with about 60% less carbon per load; team drivers cover about twice the daily distance of a solo driver at a premium of \$0.25–0.40 per mile; air at four to five times the cost of road. Last mile: the courier and local-trucking industry factors (U.S. Environmental Protection Agency 2024a) and electric-van energy use of 0.37–0.39 kWh per mile (Mulholland 2022). Facilities: site energy of 22.7 (ambient) and 84.1 (refrigerated) kBtu per square foot (U.S. Environmental Protection Agency 2024b) against a fixed operating cost of which rent at \$10.12 per square foot (Cushman & Wakefield 2025) is about 40%. Ranges combine the low and high values of every input.

Table 2: Carbon intensity per dollar of distribution activities, kg CO_2e per US dollar.

Ratio	Activity	Central	Range
Holding, r_h	Ambient packaged goods	0.077	0.030–0.223
	Bulky low-value goods	0.103	0.048–0.298
	Refrigerated dairy	0.180	0.085–0.318
	Fresh beef	0.101	0.042–0.206
	Fresh poultry	0.140	0.061–0.269
	Cold-chain pharmaceuticals	0.041	0.015–0.090
Line-haul speed, r_t	Faster vessel speed	5.37	4.97–5.82
	Truck instead of intermodal rail	1.78	1.18–3.55
	Air instead of truck	1.10	0.60–1.90
	Team instead of solo drivers	0.03	0.00–0.10
Last mile, r_L	Diesel local truck	0.50	0.30–0.60
	Diesel van	0.30	0.15–0.45
	Electric truck	0.16	0.12–0.20
	Electric van	0.075	0.05–0.10
Facilities, r_F	Ambient distribution center	0.098	0.07–0.24
	Refrigerated distribution center	0.181	0.13–0.28

is the exception: a team of two drivers runs the same tractor about twice as far per day, so the added cost is wages and the added carbon is small. Counting the carbon embodied in food that spoils while held raises the holding ratio of perishables toward the carbon intensity of the food itself, but even beef, at 1.12 kg per dollar of processed product (U.S. Environmental Protection Agency 2024a), the highest of the food categories we examine, stays below the ratio of buying speed with truck instead of intermodal rail or with vessel speed. Air freight, whose ratio spans 0.60 to 1.90, is the one fuel-bought option against which spoilage of carbon-intensive food could favor speed.

5.2 Classifying products

Proposition 7 needs, beyond these ratios, only the pooling weight ω and the speed bands, which come from the cost side of the network. In the calibrated network of Section 5.3, the cost-minimizing design operates four distribution centers, $\omega = 0.009$, and the bands are $\underline{R} = 0.61$ and $\overline{R} = 1.08$. The pooling weight is small because a distribution center's fixed cost, about \$3 million a year, is large relative to the cost of the safety stock it pools; the density condition (18) then reduces essentially to a comparison of the last mile with facilities. Table 3 reports the carbon-minimizing direction of each decision for six products, with the share of the corners of the ranges in Table 2, and of pooling weights between 0.009 and 0.6, that give the same direction.

Three patterns hold across the six products. First, speed bought with fuel is slowed under a carbon price for every product and at every corner of the ranges: the holding ratio never approaches the carbon intensity of fuel-bought speed, so every product lies below the lower band. Second, speed bought with labor is favored for most products, because its added carbon is close to zero; the direction is least settled for pharmaceuticals, whose holding ratio is the lowest. Third, the last-mile

Table 3: Carbon-minimizing direction by Proposition 7, with the percentage of range corners giving the same direction in parentheses.

	Speed bought with		Distribution centers, by last-mile vehicle			
Product	fuel	labor	diesel truck	diesel van	electric truck	electric van
Ambient packaged goods	slower (100)	faster (75)	more (98)	more (75)	more (51)	fewer (80)
Bulky low-value goods	slower (100)	faster (75)	more (96)	more (71)	fewer (54)	fewer (84)
Refrigerated dairy	slower (100)	faster (75)	more (93)	more (66)	fewer (77)	fewer (99)
Fresh beef	slower (100)	faster (75)	more (98)	more (70)	fewer (68)	fewer (98)
Fresh poultry	slower (100)	faster (75)	more (94)	more (69)	fewer (73)	fewer (99)
Cold-chain pharmaceuticals	slower (100)	faster (50)	more (100)	more (77)	fewer (60)	fewer (95)

vehicle sets the density decision: heavy diesel delivery calls for more distribution centers and electric vans for fewer, while diesel vans lean toward more and electric trucks sit close to the threshold. Within this set of products, how speed is bought and how the last mile is driven determine the direction more than the product does.

5.3 A calibrated national network

We calibrate the model to a national network that replenishes stores with packaged goods. Forty metropolitan markets each draw 250 pallets a week, with a weekly coefficient of variation of 0.40 and cross-market correlation 0.30, at a 95% cycle-service level. The source lies 1,000 miles from the distribution centers, and the distance from a center to its markets is $d(n) = 700\, n^{-1/2}$ miles, about 700 miles from a single national center and 250 from eight. A distribution center costs \$3.0 million a year to operate. Line haul is by truckload at \$2.26 per mile with 26 pallets a load, and stores are served by truck at \$2.75 per mile with 20 pallets a load. We fit $t(v)$ and $\varepsilon(v)$ through two real options on each of two lanes and let speed range between them: on the *fuel* lane, intermodal rail at 250 miles a day and truckload at 500; on the *labor* lane, a solo driver at 500 miles a day and a team at 1,000 on the same fuel. The fitted carbon rate includes a constant term, which raises reported carbon but leaves every decision unchanged because throughput is fixed. Each carbon rate is the cost rate times its ratio in Table 2. We consider two holding costs: \$390 per pallet-year for ambient packaged goods, and \$6,000 for high-value goods, about the holding cost of a pallet worth \$55,000 at the same capital charge. Both carry the ambient holding ratio of 0.077; a lower ratio for high-value goods would strengthen the slowing on the fuel lane and weaken the speeding on the labor lane.

Table 4 reports the designs. In this network, three features stand out. The first is that the last-mile vehicle sets the size of the density response. With diesel store delivery, the carbon-minimizing network has about three times as many distribution centers as the cost-minimizing one; with electric trucks it adds one or two. Electric trucks remain on the dispersal side of the threshold here because their ratio of 0.16 exceeds the facility ratio of 0.098 and the pooling weight is small.

The second is that speed responds only when holding is costly enough for speed to be worth buying. For ambient packaged goods, the cost-minimizing chain already ships at the slowest option

Table 4: Cost-minimizing and carbon-minimizing designs in the calibrated network. Changes are relative to the cost-minimizing design.

Holding cost	Speed bought	Store	Distribution centers		Line-haul days		Carbon minimum	
($/pallet-yr)	with	delivery	cost min.	carbon min.	cost min.	carbon min.	carbon	cost
390	fuel	diesel	4.1	12.2	4.0	4.0	−11.5%	+18.8%
		electric	4.1	5.7	4.0	4.0	−0.8%	+1.4%
	labor	diesel	4.1	12.2	2.0	2.0	−6.9%	+16.8%
		electric	4.1	5.7	2.0	2.0	−0.4%	+1.3%
6,000	fuel	diesel	3.8	11.5	3.4	4.0	−17.5%	+16.9%
		electric	3.8	5.3	3.4	4.0	−10.7%	+1.6%
	labor	diesel	3.9	11.8	1.8	1.1	−7.3%	+17.3%
		electric	3.9	5.5	1.8	1.2	−0.6%	+2.7%

on each lane, and the carbon-minimizing chain does too: speed bought with fuel would add carbon, and speed bought with labor saves too little safety stock to pay. For high-value goods, the cost-minimizing chain buys some speed on both lanes; carbon then returns the fuel lane to intermodal rail and moves the labor lane toward team operation, cutting line-haul time from 1.8 to about 1.1 days, as Proposition 7 predicts.

The third is that carbon prices in current use move the design little. At $100 per tonne, network carbon falls by 0.5% for ambient packaged goods with diesel delivery, and by at most 4.3% for high-value goods, because carbon is a small share of each activity's priced cost (Proposition 9); in the first case the network grows from 4.10 to 4.20 distribution centers. Reaching the carbon minimum requires a binding cap, which lowers carbon by 11.5% at an 18.8% cost increase for ambient goods with diesel delivery.

5.4 Caps on realized carbon and robustness

Under a cap on realized carbon (Proposition 10), the transport carbon rates are scaled by $m = 1 + z_{\beta_c}\mathrm{CV}_Q$. In the calibrated network $\mathrm{CV}_Q = 0.225$, so a weekly cap held with 95% confidence scales transport carbon by 1.37, and an annual cap by 1.05 because independent weekly fluctuations average out over 52 weeks. Scaling raises the line-haul and last-mile ratios, so by Corollary 11 it reinforces the directions above: slower fuel-bought speed and, with diesel delivery, more distribution centers.

The directions in Table 3 hold over the reported shares of the input ranges and of pooling weights up to 0.6. Two features of operations lie outside the model and pull in opposite directions. Pipeline inventory valued at the warehouse rates shifts the level of speed but not the speed threshold (Section 4). Vehicle fill on the last mile falls as the network disperses, which raises both the cost and the carbon of last-mile delivery per unit and pulls both optima toward consolidation (Section 4).

Table 5: Carbon-minimizing direction of the two decisions (Proposition 7), with the calibrated cases of Section 5 that fall in each cell.

	More distribution centers *last mile more carbon-intensive per dollar than facilities and lost pooling*	**Fewer distribution centers** *last mile less carbon-intensive per dollar than facilities and lost pooling*
Faster replenishment *holding more carbon-intensive per dollar than the speed it buys*	speed bought with labor; heavy diesel delivery	speed bought with labor; electric vans
Slower replenishment *holding less carbon-intensive per dollar than the speed it buys*	speed bought with fuel; heavy diesel delivery	speed bought with fuel; electric vans

6 Managerial Implications

The results give a firm a diagnostic it can compute from its own rate data, and they indicate which of its distribution decisions a carbon price or cap will move, in which direction, and by how much. Each implication below rests on a result of Section 4; the calibrated values of Section 5 illustrate it.

A diagnostic in two questions. A firm needs four ratios and one cost share. The ratios are the carbon per dollar of holding stock, of the speed it can buy on its line-haul lanes, of last-mile delivery, and of operating a distribution center. Each is close to the activity's energy share of cost times the carbon of a dollar of its energy (Section 5.1), so it can be estimated from fuel and electricity spend. The cost share is the pooling weight ω, the share of a distribution center's marginal cost that is lost pooling, which the firm can read from its cost-minimizing network through (16). Proposition 7 then answers two questions: whether replenishment should be faster or slower, and whether the network should have more or fewer distribution centers. Table 5 sets out the four answers.

Slow the speed that is bought with fuel. A faster vessel, truck instead of intermodal rail, and air instead of truck add cost that is mostly fuel, so their carbon per dollar is many times that of holding stock. Proposition 3 then calls for slower replenishment, and in our calibration it does so for every product, perishables and pharmaceuticals included (Table 3). The shifts of pharmaceutical freight from air to ocean and of truckload lanes to intermodal rail are consistent with this direction (Pharmaceutical Commerce 2023; FreightWaves 2026). Where the cost-minimizing chain already ships at the slowest option, as it does for ambient goods in Section 5.3, a carbon price leaves speed there.

Buy speed with labor where speed pays. Speed bought with labor, such as a team of drivers on the same tractor, adds wages rather than fuel, so its carbon per dollar is close to zero and a

carbon price favors it (Proposition 3). It matters where holding is costly enough for speed to pay at all: in Section 5.3, carbon moves the labor lane toward team operation for high-value goods and leaves ambient goods with a solo driver. Demand volatility and the service target raise the cost-minimizing and the carbon-minimizing speeds together, so they change how fast the chain runs but not the direction in which carbon moves it.

Let the last-mile vehicle set the number of distribution centers. The density decision compares the carbon per dollar of last-mile delivery with that of a distribution center's facility and lost pooling (Proposition 7). Heavy diesel delivery is carbon-intensive enough to call for more distribution centers closer to demand, about three times as many at the carbon minimum of Section 5.3; electric vans call for fewer; diesel vans and electric trucks lie close to the threshold, where facility energy and the pooling weight decide. A firm that electrifies its last mile should therefore revisit how many distribution centers it operates, and one that adds centers to shorten delivery should weigh the vehicle fill it gives up, since emptier vehicles raise both the cost and the carbon of each delivered unit (Section 4).

Plan line haul and network together. Because faster replenishment recovers part of the pooling that dispersion gives up, the carbon intensity of line-haul freight enters the density condition (Proposition 7): a lower-emission line haul lowers the threshold for adding distribution centers. The effect works through the pooling weight, so it is small for networks whose fixed costs are large relative to their safety stock, as in Section 5.3, and larger for volatile products and high service targets, for which lost pooling is a larger share of a center's marginal cost (Proposition 4). More generally, the two decisions are complements (Proposition 6), so a program that acts on one alone, such as a slow-steaming mandate or a consolidation drive, evaluates it at a level of the other decision that the joint design would change.

Expect modest moves at current carbon prices. Proposition 9 ties the response to a carbon price to the share of carbon in each activity's priced cost. At \$100 per tonne, carbon is about 1% of the priced cost of holding and of facilities, 5% of diesel truck delivery, 15% of speed bought with truck instead of rail, and 35% of speed bought with vessel speed; it reaches half of the priced cost of diesel truck delivery only at about \$2,000 per tonne. A carbon price in the range now in use therefore moves the design little, by less than one percent of network carbon for ambient goods in Section 5.3, and it moves fuel-bought speed first, which is consistent with the attention slow steaming has received as a carbon response in shipping (Psaraftis and Kontovas 2013). Larger redesigns follow from caps that bind: at the carbon minimum the network of Section 5.3 has about three times as many distribution centers. Without pooling, the carbon that a cost-minimizing design leaves unabated grows, to second order, with the square of the logarithm of each ratio (Corollary 8), so redesign pays most for products whose ratios lie far from one.

Account for realized carbon over short horizons. A cap on realized rather than expected carbon places a risk margin on transport emissions (Proposition 10). For a weekly cap held with 95% confidence, the margin raises the transport carbon rates by 37% in our calibration, which reinforces slower fuel-bought speed and, with diesel delivery, more distribution centers (Corollary 11); for an annual cap the margin is about 5% and changes little.

Implications for policy. An instrument that prices or caps transport alone, such as a freight fuel levy or a vehicle emissions standard, blends the carbon price into the transport rates and leaves holding and facility carbon unpriced. At a given density it therefore slows replenishment, and at a given speed it adds distribution centers, whatever the product. By the marginal comparisons of Propositions 3 and 5, these moves lower total carbon when they agree with the carbon-minimizing direction and, for small prices, raise it when they do not: a transport-only price raises total carbon at the cost optimum when holding is more carbon-intensive per dollar than the speed it would buy, or when the last mile is less carbon-intensive per dollar than facilities and lost pooling. In our calibration the first case arises when speed bought with labor is in use and the second with electric last-mile vans. An instrument that prices total logistics carbon, including warehousing, lets the firm choose the speed and density that minimize carbon across all four activities.

7 Discussion and Conclusion

We have studied a distribution network that provides two services, availability and proximity, each of which the firm can provide through more than one operational activity: safety stock or replenishment speed for availability, facilities or last-mile distance for proximity. The carbon-minimizing configuration depends on which activity provides each service at lower carbon intensity per dollar. The contribution of the paper is to make these two substitutions explicit and to give the thresholds at which the carbon-minimizing configuration departs from the cost-minimizing one.

Limitations. The thresholds apply to an average, symmetric distribution environment, so their use in a heterogeneous network is diagnostic rather than mechanical. Heterogeneous markets or discrete facility sites replace $d(n) = \delta n^{-1/2}$ with the realized last-mile profile; this can move the density threshold, but it leaves the underlying comparison between facility carbon per dollar and last-mile carbon per dollar unchanged. The representative inbound distance isolates the proximity effect of adding DCs. If density also changes inbound routing, shipment consolidation, or facility scale economies, those mechanisms add centralizing forces and can move a near-boundary case toward consolidation. The base model keeps last-mile rates fill-independent; the load-factor sensitivity of Section 4 shows how this assumption matters: common fill penalties consolidate both optima, while fill penalties that fall more heavily on carbon can reverse the density comparison. Service enters as a cycle-service requirement; if service or fill rate is chosen endogenously, safety stock becomes an additional design margin and the speed threshold should be evaluated at the chosen service level. Finally, the calibration is regime-representative rather than firm-estimated,

so the numerical values illustrate the order of magnitude of the threshold movements under documented rate ranges. The thresholds are therefore diagnostics to be evaluated with product- and network-specific rate data.

Future research. The most direct extensions are those that can move the threshold boundaries. Multi-product and multi-echelon settings would show how a shared network accommodates products in different regimes and where safety stock should be held. Explicit load-factor, shipment-consolidation, and routing models would quantify the centralizing force represented here by the load sensitivity. Heterogeneous geography and discrete sites would replace the continuous density curve with network-specific last-mile profiles. Empirical calibration with firm-level logistics-emissions data would estimate where real products fall relative to the speed and density boundaries, and endogenous service choice would make the cycle-service target a design decision rather than an input.

The model identifies which operational activity supplies availability and which supplies proximity at lower carbon intensity per dollar, and shows how the two substitutions interact in the joint distribution design: the speed threshold prices the availability margin, the pooling-adjusted density threshold prices the proximity margin, and complementarity links the two in the final configuration.

A Proofs

Proof of Lemma 1. The blended rates in (8) follow by collecting coefficients in $K + \tau C$. The speed comparison sets holding against inbound transport at the margin (Proposition 3); the density comparison sets last-mile transport against a pooling-adjusted blend of facilities and holding (Proposition 5). The four ratios listed in the statement are therefore the primitive carbon-intensity comparisons used below. □

Proof of Proposition 2. We give the argument for J; for C replace each blended rate by its carbon rate. (i) With $k = z\sigma\sqrt{ND}$, the holding term $H_\tau \mathrm{SS}(n, v) = H_\tau k\sqrt{\Psi(n)}\, v^{-1/2}$ is strictly convex in v and T_τ is convex, so the minimizer over $[v_{\min}, v_{\max}]$ is unique and, by the maximum theorem, continuous in n. It depends on n only through $\Psi(n)$: the marginal holding saving $\frac{1}{2}H_\tau k\sqrt{\Psi}\, v^{-3/2}$ rises in Ψ, while the marginal transport cost $N\mu D\, T'_\tau(v)$ does not depend on it, so v^* is nondecreasing in Ψ and therefore in n. (ii) By the envelope theorem in the form of Danskin's theorem, which applies because the speed response is unique, J^{r} is differentiable with $J^{\mathrm{r}}_n(n) = J_n\big(n, v^*(n)\big)$. Multiplying by $n^{3/2} > 0$ preserves its sign:

$$n^{3/2} J^{\mathrm{r}}_n(n) = F_\tau\, n^{3/2} + H_\tau\, n^{3/2} G\big(n, v^*(n)\big) - \tfrac{1}{2}\,\ell_\tau N\mu\delta,$$

with $G = \partial \mathrm{SS}/\partial n$ as in (9). The first term is strictly increasing. Where v^* is interior, (10) gives

$H_\tau k\, v^{-1/2} = 2N\mu D\, T'_\tau(v)\, v/\sqrt{\Psi}$, so

$$H_\tau\, n^{3/2} G\big(n, v^*(n)\big) = (1-\rho)\, N\mu D\, \frac{n^{3/2}}{\Psi(n)}\, T'_\tau\big(v^*(n)\big)\, v^*(n).$$

The factor $n^{3/2}/\Psi(n)$ is strictly increasing, since its elasticity in n is $\frac{3}{2} - n(1-\rho)/\Psi(n) \geq \frac{1}{2}$, and $T'_\tau(v)\, v$ is nonnegative and nondecreasing in v by convexity, hence nondecreasing in n by (i). Where a speed bound binds, $H_\tau n^{3/2} G \propto n^{3/2}\Psi(n)^{-1/2}$, which is also strictly increasing. The middle term is continuous in n, so $n^{3/2} J^{\mathrm{r}}_n$ is strictly increasing on $[1, N]$ and changes sign at most once, from negative to positive. At a fixed speed the same argument applies with v constant. (iii) The minimizer of J^{r} over $[1, N]$ is therefore unique, and with (i) so is the joint minimizer. Since J is continuous in (n, v, τ) and the minimizer over the compact feasible set is unique, it is continuous in τ by the maximum theorem. □

Proof of Proposition 3. $N\mu D\, t(v)$ and $N\mu D\, \varepsilon(v)$ are convex and $\kappa(n) v^{-1/2}$ has $\partial^2_v = \frac{3}{4}\kappa(n) v^{-5/2} > 0$, so both objectives are strictly convex in v. The cost optimum satisfies $\hat{K}'(v_{\mathrm{K}}) = 0$, i.e. $N\mu D\, t'(v_{\mathrm{K}}) = \frac{1}{2} h\, \kappa(n)\, v_{\mathrm{K}}^{-3/2}$, so that $\frac{1}{2}\kappa(n) v_{\mathrm{K}}^{-3/2} = N\mu D\, t'(v_{\mathrm{K}})/h$. Evaluating the carbon derivative there,

$$\hat{C}'(v_{\mathrm{K}}) = -\eta\, \tfrac{1}{2}\kappa(n) v_{\mathrm{K}}^{-3/2} + N\mu D\, \varepsilon'(v_{\mathrm{K}}) = N\mu D\big(\varepsilon'(v_{\mathrm{K}}) - (\eta/h)\, t'(v_{\mathrm{K}})\big),$$

which is negative — so $v_C > v_{\mathrm{K}}$ by strict convexity — if and only if (11). With $t' = 2t_1 v$ and $\varepsilon' = 2\varepsilon_0 v$ the marginal ratio is ε_0/t_1, and the priced first-order condition (10) gives $v^{5/2} = (h + \tau\eta) z\sigma\sqrt{ND}\,\sqrt{\Psi}/[4(t_1 + \tau\varepsilon_0) N\mu D]$, which is (12). The projection and the boundary qualification are immediate. □

Proof of Proposition 4. Let $s = z\sigma$. From (12), $v^*(n; s) \propto s^{2/5}$, so speed rises with service and demand volatility. Along this speed response, $J^{\mathrm{r}}_n(n; s) = J_n\big(n, v^*(n; s); s\big)$ because $J_v = 0$. The holding marginal term in (9) is proportional to $s(v^*)^{-1/2}(1-\rho)/\sqrt{\Psi(n)} \propto s^{4/5}(1-\rho)/\sqrt{\Psi(n)}$, hence rises in s even after the induced speed increase. Thus $J^{\mathrm{r}}_{ns} > 0$. By Proposition 2, $n^{3/2} J^{\mathrm{r}}_n(n; s)$ is strictly increasing in n, and it is increasing in s; its single sign change therefore occurs at a lower n when s rises, so the optimal density is nonincreasing in s, and strictly decreasing where it is interior. □

Proof of Proposition 5. By Proposition 2(ii), $J(\cdot, v)$ and $C(\cdot, v)$ are strictly quasiconvex on $[1, N]$, which gives uniqueness and the interior/boundary characterization. For the threshold, take $\tau = 0$ for cost and $\tau \to \infty$ for carbon. The cost density satisfies $\partial_n K = F + h\, G(n_K, v) - t_L N\mu\delta/(2n_K^{3/2}) = 0$, so the last-mile marginal $t_L N\mu\delta/(2n_K^{3/2}) = F + h\, G(n_K, v)$. Evaluating the carbon derivative there,

$$\partial_n C\big|_{n_K} = \varphi + \eta\, G(n_K, v) - \frac{\varepsilon_L N\mu\delta}{2n_K^{3/2}} = \varphi + \eta\, G(n_K, v) - \frac{\varepsilon_L}{t_L}\big[F + h\, G(n_K, v)\big],$$

which is negative — so $n_C > n_K$ by strict quasiconvexity of $C(\cdot, v)$ — if and only if (14). Writing

$R(G) = (\varphi + \eta G)/(F + hG)$, we have $R(0) = \varphi/F$, $R(G) \to \eta/h$ as $G \to \infty$, and $R'(G) = (\eta F - h\varphi)/(F + hG)^2$, so R is monotone with $\operatorname{sign} R' = \operatorname{sign}(\eta/h - \varphi/F)$ and lies between φ/F and η/h. In the no-pooling limit $\rho \to 1$, $G \to 0$ and $\Psi \to N$, so (9) gives $n^* = (\ell_\tau N \mu\delta/2F_\tau)^{2/3}$ and (14) becomes (15). □

Proof of Proposition 6. Since $G(n, v) \propto v^{-1/2}$, the cross-partial is $J_{nv} = H_\tau G_v$, which equals $-H_\tau G(n, v)/(2v) < 0$ for $\rho < 1$. For the speed response, where v^* is interior (10) holds; its right side strictly increases in n and strictly decreases in v, while its left side does not depend on n and is nondecreasing in v, so v^* strictly increases in n, and projection onto the speed bounds preserves monotonicity. For the density response, $n^{3/2} J_n(n, v)$ is strictly increasing in n (Proposition 2) and, since $J_{nv} < 0$, decreasing in v; its single sign change therefore occurs at a higher n when v rises. □

Proof of Proposition 7. (i) By Proposition 2 applied to the carbon objective, $n_C > n_K$ if and only if the reduced carbon objective slopes downward at n_K, $C^{\mathrm{r}}_n(n_K) < 0$, and $n_C = n_K$ if it is zero. By the envelope theorem, $C^{\mathrm{r}}_n(n_K) = \varphi + \eta\, G\big(n_K, v_C(n_K)\big) - \frac{1}{2}\varepsilon_L N \mu\delta\, n_K^{-3/2}$, where $v_C(n_K)$ is the carbon-minimizing speed at density n_K. The cost condition (9) at $\tau = 0$ gives $\frac{1}{2} t_L N \mu\delta\, n_K^{-3/2} = F + hG(n_K, v_K)$, so the last term equals $r_L\big[F + hG(n_K, v_K)\big]$. By (12), at a common density the carbon- and cost-minimizing speeds stand in the ratio $R^{2/5}_{\mathrm{speed}}$, and $G \propto v^{-1/2}$, so $\eta\, G\big(n_K, v_C(n_K)\big) = \eta\, G(n_K, v_K)\, R^{-1/5}_{\mathrm{speed}} = h\, G(n_K, v_K)\, r_h^{4/5} r_t^{1/5}$. Dividing $C^{\mathrm{r}}_n(n_K)$ by $F + hG(n_K, v_K) > 0$ and writing $\varphi = F\, r_F$ gives (18). The second expression for ω in (16) follows from $F + hG(n_K, v_K) = \frac{1}{2} t_L N \mu\delta\, n_K^{-3/2}$ and $F = \frac{1}{2} t_L N \mu\delta\, (n^0_K)^{-3/2}$. (ii) By (12), $v^*(n; \tau)$ equals $\big[(h + \tau\eta)/(t_1 + \tau\varepsilon_0)\big]^{2/5} \Psi(n)^{1/5}$ times a factor that depends on neither n nor τ; evaluating at $(n_C, \tau \to \infty)$ and $(n_K, 0)$ and taking logarithms gives (19). (iii) If (18) holds, then $n_C > n_K$ and $\Psi(n_C) > \Psi(n_K)$, so in (19) the first term is nonnegative and the second positive; the reversed case is symmetric. Since Ψ is increasing with $\Psi(1) \le \Psi(n_C) \le \Psi(N) = N$, the threshold $\sqrt{\Psi(n_K)/\Psi(n_C)}$ in (ii) lies in $[\underline{R}, \overline{R}]$. □

Proof of Corollary 8. With $\rho = 1$, $\Psi \equiv N$ and $G \equiv 0$, so (9) involves only F_τ and ℓ_τ and gives $n^*(\tau) = (\ell_\tau N \mu\delta/2F_\tau)^{2/3}$ (Proposition 5); the ratio of its carbon and cost values is $(r_L/r_F)^{2/3}$. The speed response (12) no longer depends on n, and the ratio of its carbon and cost values is $R^{2/5}_{\mathrm{speed}}$. For (20), a function $an + bn^{-1/2}$ attains its minimum $3an^\circ$ at $n^\circ = (b/2a)^{2/3}$ and equals $an^\circ(x + 2x^{-1/2})$ at $n = xn^\circ$; with $n_K = xn_C$, $P(n_K)/P(n_C) = (x + 2x^{-1/2})/3$. Likewise $av^{-1/2} + bv^2$ attains its minimum $\frac{5}{4} a (v^\circ)^{-1/2}$ and equals $a(v^\circ)^{-1/2}(y^{-1/2} + y^2/4)$ at $v = yv^\circ$, so with $v_K = yv_C$, $A(v_K)/A(v_C) = (4y^{-1/2} + y^2)/5$. Weighting by θ and $1 - \theta$ gives (20); expanding $x = e^{u_1}$ and $y = e^{u_2}$ to second order gives the last statement. □

Proof of Proposition 9. (i) By (12), $\ln v^* = \frac{2}{5}\big[\ln(h + \tau\eta) - \ln(t_1 + \tau\varepsilon_0)\big]$ plus terms that do not depend on τ, and $\mathrm{d}\ln(h + \tau\eta)/\mathrm{d}\ln\tau = s_h$. (ii) The same computation applies to $n^* = (\ell_\tau N \mu\delta/2F_\tau)^{2/3}$. (iii) By (12), along the speed response $H_\tau G\big(n, v^*(n)\big) = c\,(1 - \rho)\, H_\tau^{4/5} b_\tau^{1/5} \Psi(n)^{-3/5}$, with $b_\tau =$

$t_1 + \tau\varepsilon_0$ and $c > 0$ depending on neither n nor τ. Dividing the expression for $n^{3/2}J_n^{\mathrm{r}}$ in the proof of Proposition 2 by $\ell_\tau > 0$,

$$\frac{n^{3/2}J_n^{\mathrm{r}}}{\ell_\tau} = \frac{F_\tau}{\ell_\tau}\, n^{3/2} + c\,(1-\rho)\,\frac{H_\tau^{4/5} b_\tau^{1/5}}{\ell_\tau}\,\frac{n^{3/2}}{\Psi(n)^{3/5}} - \tfrac{1}{2}N\mu\delta.$$

If both coefficients are nonincreasing in τ, the expression is pointwise nonincreasing in τ, so its single sign change moves weakly to higher n and $n^*(\tau)$ is nondecreasing. The logarithmic derivatives of the coefficients are $1/(r_F^{-1}+\tau) - 1/(r_L^{-1}+\tau)$ and $\frac{4}{5}/(r_h^{-1}+\tau) + \frac{1}{5}/(r_t^{-1}+\tau) - 1/(r_L^{-1}+\tau)$. Since $1/(r^{-1}+\tau)$ is increasing in r, both are nonpositive at every τ when r_L exceeds r_F, r_h, and r_t, and nonnegative when it falls short of each. □

Proof of Proposition 10. In (21) the throughput coefficient $\gamma(n,v) = \varepsilon(v)D + \varepsilon_L d(n) > 0$, so $\{\tilde{C} \le B\} = \{Q \le (B - \varphi n - \eta\mathrm{SS})/\gamma\}$, and $\Pr\{\tilde{C} \le B\} \ge 1-\beta_c$ holds iff $(B - \varphi n - \eta\mathrm{SS})/\gamma \ge q_{1-\beta_c}$, i.e. $\varphi n + \eta\mathrm{SS} + \gamma\, q_{1-\beta_c} \le B$. Writing $\gamma\, q_{1-\beta_c} = (q_{1-\beta_c}/N\mu)\,[\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)]$ gives (22). Under normality $q_{1-\beta_c} = N\mu + z_{\beta_c}\sigma_Q$, so $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$. □

Proof of Corollary 11. The chance constraint multiplies the inbound and last-mile carbon rates by m (Proposition 10); substituting $m\varepsilon_0$ and $m\varepsilon_L$ for ε_0 and ε_L in (11) and (14) gives the two conditions, the latter scaling only the left side and leaving the pooling-adjusted facility threshold $(\varphi+\eta G)/(F+hG)$ on the right. The same substitution in Proposition 7, with n_K and ω unchanged because the cost rates are unchanged, gives the joint conditions. From $\sigma_Q^2 = \sigma^2 N[1+(N-1)\rho]$, $\mathrm{CV}_Q = (\sigma/\mu)\sqrt{[1+(N-1)\rho]/N}$ is increasing in σ/μ and, for $N > 1$, in ρ, and z_{β_c} is increasing in $1-\beta_c$, so $m = 1 + z_{\beta_c}\mathrm{CV}_Q$ is increasing in all three. For the joint speed, divide the carbon density condition by m: its facility and lost-pooling terms then scale as m^{-1} and $m^{-4/5}$ and, multiplied by $n^{3/2}$, rise in n with elasticities $\frac{3}{2}$ and at least $\frac{9}{10}$, so that, at an interior n_C, $\mathrm{d}\ln n_C/\mathrm{d}\ln m \le \frac{8}{9}$. Since Ψ has elasticity at most one in n, (19) with R_{speed}/m in place of R_{speed} gives $\mathrm{d}\ln v_C/\mathrm{d}\ln m \le -\frac{2}{5} + \frac{1}{5}\cdot\frac{8}{9} < 0$. □

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